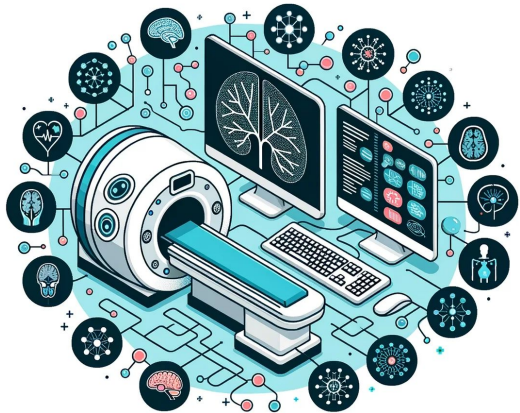




Drug Discovery using Deep Learning

Session 09

Nov 01 2025

The Drug Discovery Challenge

The Central Paradigm

Sequence



Structure



Function

Traditional Approaches

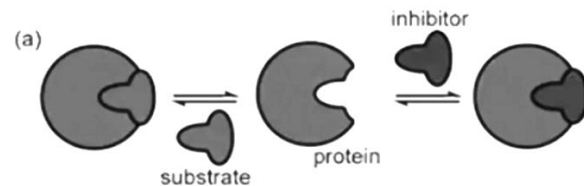
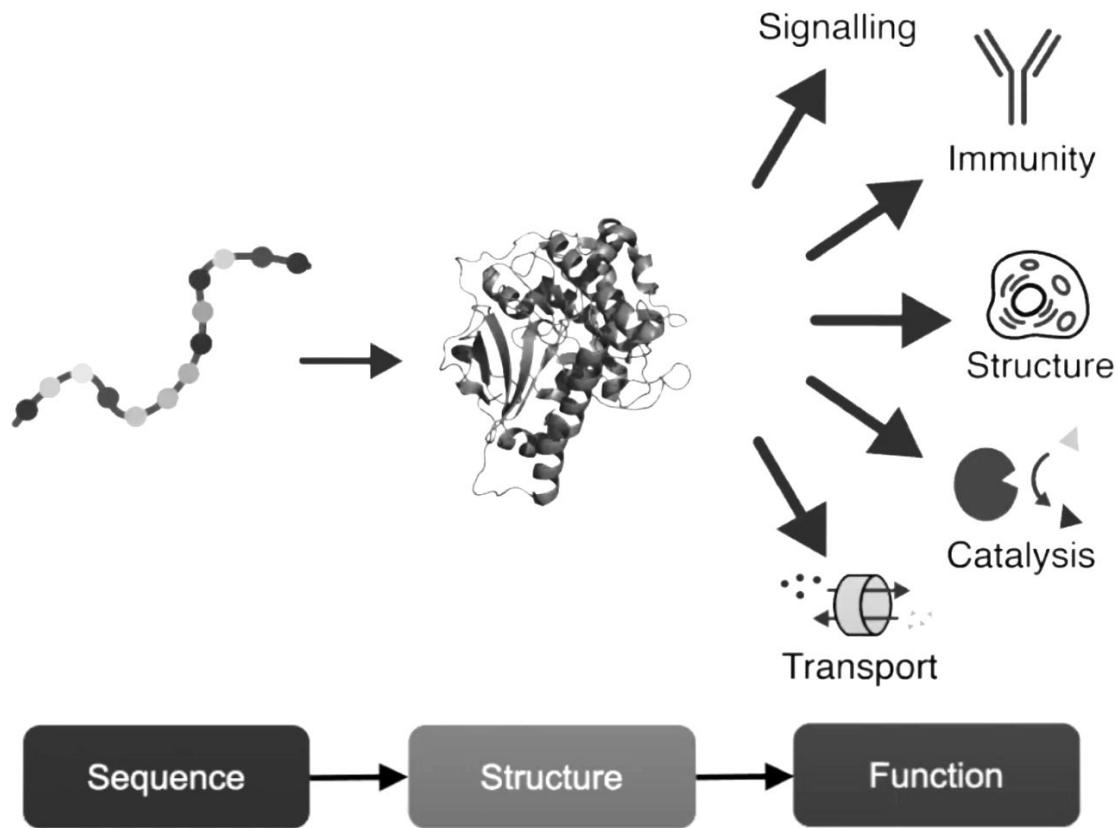
- High-throughput screening
- Structure-based design
- Fragment-based discovery
- Time: 10-15 years
- Cost: \$2.6B per drug

AI-Driven Discovery

- Generative models
- Virtual screening
- De novo design
- Time: 2-3 years (potential)
- Cost: Significantly reduced

Key Insight: Deep learning enables us to navigate chemical space more efficiently, generating novel molecules with desired properties while respecting physical and chemical constraints.

The Drug Discovery Challenge



Wilson, A. J. (2009). Chemical Society Reviews, 38(12), 3289-3300.

Modern Drug Design Strategies

1. Fragment-Based Drug Design

Building larger molecules from smaller fragments that bind to different sites of the target protein

Process: Fragment screening → Hit identification → Fragment growing/merging/linking → Lead optimization

2. Scaffold Hopping

Replacing the core structure of a molecule while maintaining biological activity

Advantage: Improves pharmacokinetics, avoids patents, reduces toxicity

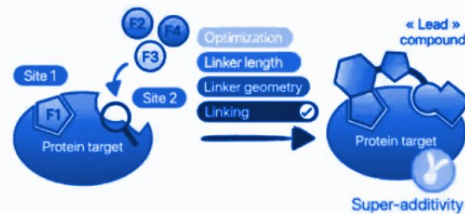
3. Targeted Protein Degradation

Using PROTACs and molecular glues to induce protein degradation via the ubiquitin-proteasome system

Innovation: Targets previously "undruggable" proteins by degradation rather than inhibition

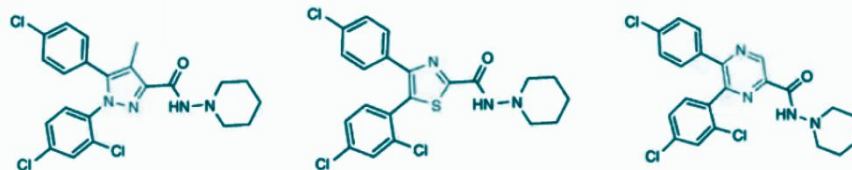
Modern Drug Design Strategies

1. Fragment-based drug design [Bancet A. et al., 2020]

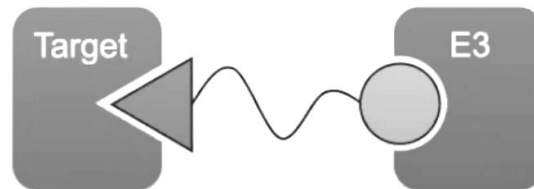


- Molecule growing
- Fragment merging
- Fragment linking

2. Scaffold hopping [Sun H. et al., 2012]



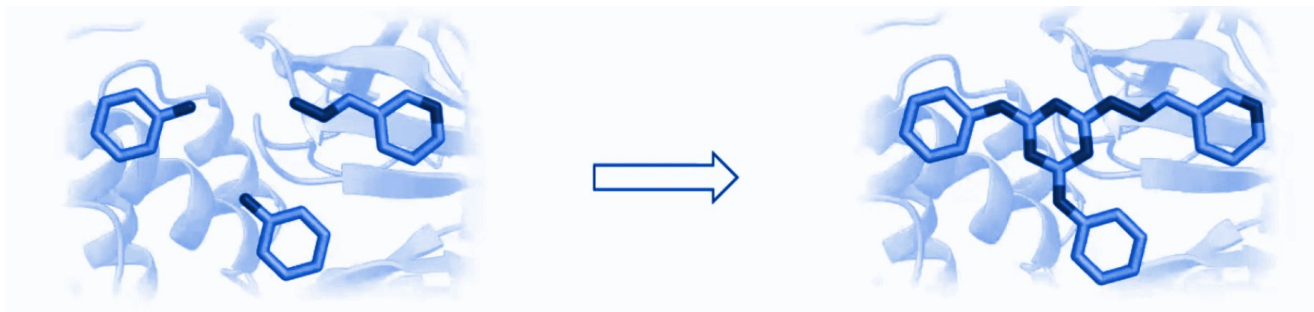
3. Targeted protein degradation [Békés M. et al., 2022]



- PROTACs
- Molecular Glues

Problem Setup

- Fragments placed in 3D space (known fixed orientation)
- Variable linker size
- Unknown attachment atoms (anchors)
- Variable number of fragments
- No clashes with protein pocket



Diffusion Models: Theoretical Foundation

Core Concept

Diffusion models learn to generate data by reversing a gradual noising process. For molecules, this means learning to construct valid molecular structures from random noise.

Forward Process (Fixed)

$$q(z_t | z_{t-1}) = N(z_t; \sqrt{\alpha_t} z_{t-1}, \sigma_t^2 I)$$

Gradually add Gaussian noise to the molecule representation over T timesteps

Reverse Process (Learned)

$$p(z_{t-1} | z_t) = q(z_{t-1} | \hat{x}, z_t)$$

Neural network learns to denoise, predicting the original molecule at each step

Training Objective

$$\mathbb{E}_{t \sim U(0, \dots, T)} [T \cdot ||\epsilon - \hat{\epsilon}_t||^2]$$

Minimize the difference between predicted and actual noise across all timesteps

Diffusion Models: Drug Design

Forward diffusion process:

$$q(z_t|z_{t-1}) = \mathcal{N}(z_t; \bar{\alpha}_t z_{t-1}, \bar{\sigma}_t^2 I)$$

$$q(z_0, z_1, \dots, z_T|x) = q(z_0|x) \prod_{t=1}^T q(z_t|z_{t-1})$$

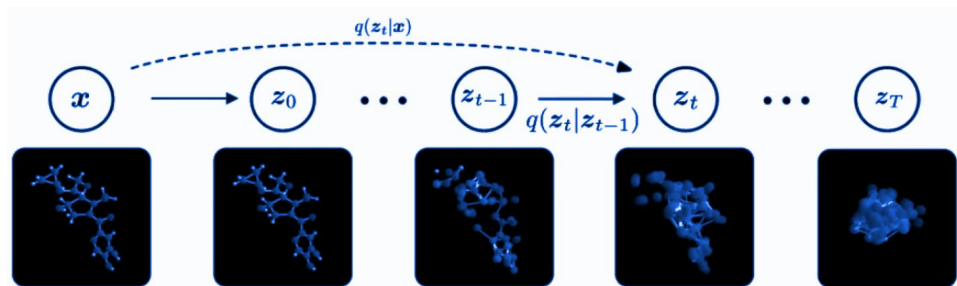
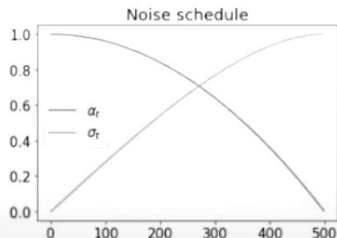
$$q(z_t|x) = \mathcal{N}(z_t|\alpha_t x, \sigma_t^2 I)$$

Noise parameters and schedule:

$$\bar{\alpha}_t = \alpha_t / \alpha_{t-1} \quad \alpha_t \propto 1 - (t/T)^2$$

$$\bar{\sigma}_t^2 = \sigma_t^2 - \bar{\alpha}_t^2 \sigma_{t-1}^2$$

$$\alpha_t = \sqrt{1 - \sigma_t^2}$$



- coordinates x_r
- atom types x_h

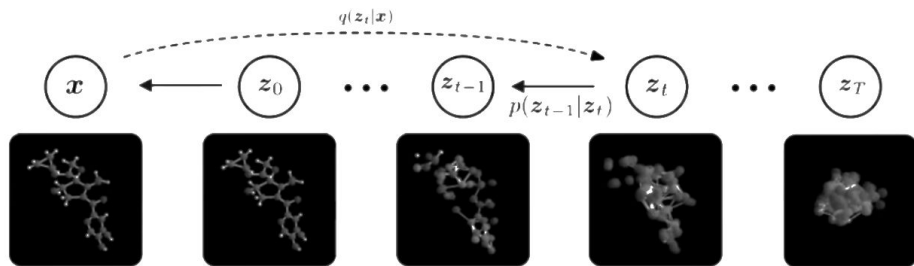
$$x = [x_r, x_h]$$

Diffusion Models: Drug Design

Forward diffusion process:

$$q(\mathbf{z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t; \bar{\alpha}_t \mathbf{z}_{t-1}, \bar{\sigma}_t^2 \mathbf{I})$$

$$q(\mathbf{z}_t | \mathbf{x}) = \mathcal{N}(\mathbf{z}_t | \alpha_t \mathbf{x}, \sigma_t^2 \mathbf{I})$$



True denoising process:

$$q(\mathbf{z}_{t-1} | \mathbf{x}, \mathbf{z}_t) = \mathcal{N}(\mathbf{z}_{t-1}; \boldsymbol{\mu}_t(\mathbf{x}, \mathbf{z}_t), \varsigma_t^2 \mathbf{I}), \quad \boldsymbol{\mu}_t(\mathbf{x}, \mathbf{z}_t) = \frac{\bar{\alpha}_t \sigma_{t-1}^2}{\sigma_t^2} \mathbf{z}_t + \frac{\alpha_s \bar{\sigma}_t^2}{\sigma_t^2} \mathbf{x} \quad \text{and} \quad \varsigma_t = \frac{\bar{\sigma}_t \sigma_{t-1}}{\sigma_t}$$



Generative denoising process: Instead of true $\mathbf{x}$ we use some approximation $\hat{\mathbf{x}}$

$$p(\mathbf{z}_{t-1} | \mathbf{z}_t) = q(\mathbf{z}_{t-1} | \hat{\mathbf{x}}, \mathbf{z}_t), \quad \text{where} \quad \hat{\mathbf{x}} = (1/\alpha_t) \mathbf{z}_t - (\sigma_t/\alpha_t) \hat{\boldsymbol{\epsilon}}_t$$

$$\hat{\boldsymbol{\epsilon}}_t = \varphi(\mathbf{z}_t, t)$$



Learning objective: $\mathbb{E}_{t \sim \mathcal{U}(0, \dots, T)} [T \cdot \|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}_t\|^2]$

Equivariance in Molecular Systems

What is Equivariance?

A function f is equivariant to transformations if transforming the input and then applying f gives the same result as applying f and then transforming the output.

$$f(T[g] \circ x) = T[g] \circ f(x)$$

Where $T[g]$ is a transformation (rotation, translation) and g is from group G

Why It Matters for Molecules

- Molecular properties are invariant to rotation/translation
- Binding affinity doesn't change with orientation
- Physical laws respect these symmetries
- Reduces sample complexity

SE(3) Equivariance

- SE(3): Special Euclidean group in 3D
- Covers rotations and translations
- O(3): Rotations only
- Essential for 3D molecular generation

Practical Implementation

Use geometric features: interatomic distances, angles, and relative positions rather than absolute coordinates. Process through equivariant graph neural networks (EGNNs).

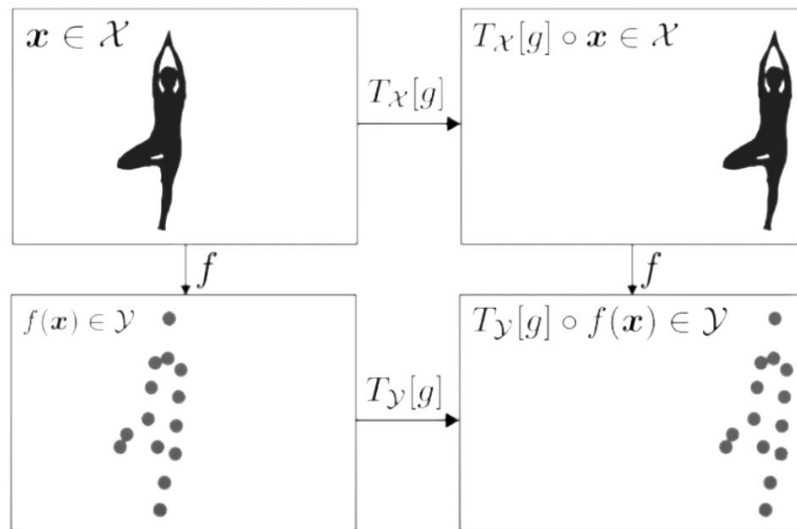
Equivariance

Function $f : \mathcal{X} \rightarrow \mathcal{Y}$ is equivariant w.r.t. group $\mathcal{G}$ if

$$f(T_{\mathcal{X}}[g] \circ \mathbf{x}) = T_{\mathcal{Y}}[g] \circ f(\mathbf{x}) \quad \forall g \in \mathcal{G} \quad \forall \mathbf{x} \in \mathcal{X}$$

Distribution $p(\mathbf{x}|\mathbf{y})$ is equivariant w.r.t. group $\mathcal{G}$ if

$$p(T_{\mathcal{X}}[g] \circ \mathbf{x} | T_{\mathcal{Y}}[g] \circ \mathbf{y}) = p(\mathbf{x} | \mathbf{y}) \\ \forall g \in \mathcal{G}, \forall \mathbf{x} \in \mathcal{X}, \forall \mathbf{y} \in \mathcal{Y}$$



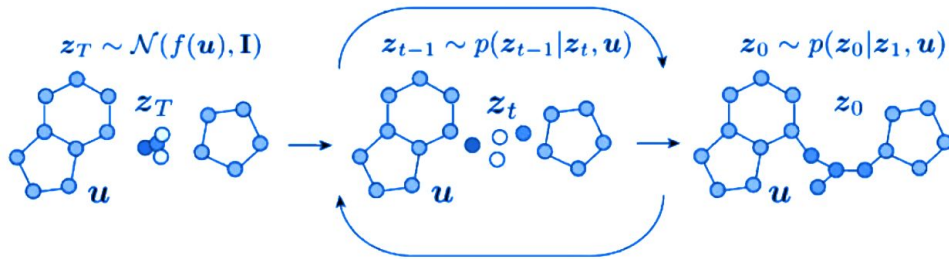
Worrall, D. E. et al. (2017) CVPR

3D Conditioning

Additional fixed 3D context $\mathbf{u}$:

$$\mathbf{x} \rightarrow [\mathbf{x}, \mathbf{u}]$$

$$\mathbf{z}_t \rightarrow [\mathbf{z}_t, \mathbf{u}]$$



$$p(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{u}) = q(\mathbf{z}_{t-1} | \hat{\mathbf{x}}, \mathbf{z}_t), \quad \text{where} \quad \hat{\mathbf{x}} = (1/\alpha_t)\mathbf{z}_t - (\sigma_t/\alpha_t)\varphi(\mathbf{z}_t, \mathbf{u}, t)$$

Proposition 1 Consider a prior noise distribution $p(\mathbf{z}_T | \mathbf{u}) = \mathcal{N}(\mathbf{z}_T; f(\mathbf{u}), \mathbf{I})$ and transition distributions $p(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{u}) = q(\mathbf{z}_{t-1} | \hat{\mathbf{x}}, \mathbf{z}_t)$, where q is an isotropic Gaussian, $f: \mathbb{R}^{M \times 3} \rightarrow \mathbb{R}^3$ is a function operating on 3D point clouds, and $\hat{\mathbf{x}}$ is an approximation computed by the neural network φ . Let the conditional denoising probabilistic model p be a Markov chain defined as

$$p(\mathbf{z}_0, \mathbf{z}_1, \dots, \mathbf{z}_T | \mathbf{u}) = p(\mathbf{z}_T | \mathbf{u}) \prod_{t=1}^T p(\mathbf{z}_{t-1} | \mathbf{z}_t, \mathbf{u}). \quad (9)$$

If f is $O(3)$ -equivariant and φ is equivariant to joint $O(3)$ -transformations of $\mathbf{z}_t$ and $\mathbf{u}$, then $p(\mathbf{z}_0 | \mathbf{u})$ is $O(3)$ -equivariant.

Equivariant Graph Neural Networks

1. Message Computation

$$m_{ij} = \phi_e(h_i, h_j, d^2_{ij}, a_{ij})$$

Compute messages between atoms i and j using node features h , squared distances d^2 , and edge attributes a

2. Feature Update

$$h_i^{l+1} = \phi_h(h_i^l, \sum_{j \neq i} m_{ij})$$

Aggregate messages to update node features (invariant properties like atom type, charge)

3. Coordinate Update

$$r_i^{l+1} = r_i^l + \sum_{j \neq i} (r_i - r_j) / d_{ij+1} \cdot \phi_r(h_i, h_j, d^2_{ij})$$

Update 3D coordinates equivariantly using relative position vectors

Equivariant Graph Neural Network

- Process coordinates and features separately
- Compute interatomic distances:

$$d_{ij} = \|\mathbf{r}_i^l - \mathbf{r}_j^l\|$$

- Message:

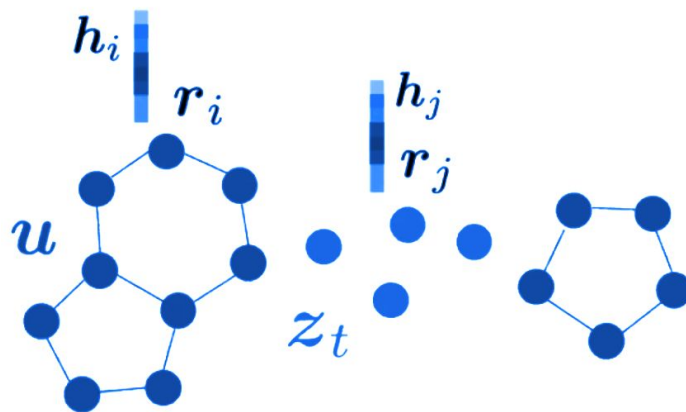
$$\mathbf{m}_{ij} = \phi_e(\mathbf{h}_i^l, \mathbf{h}_j^l, d_{ij}^2, a_{ij})$$

$$\tilde{e}_{ij} = \phi_{inf}(\mathbf{m}_{ij})$$

- Update:

$$\mathbf{h}_i^{l+1} = \phi_h(\mathbf{h}_i^l, \sum_{j \neq i} \tilde{e}_{ij} \mathbf{m}_{ij}) \quad \mathbf{r}_i^{l+1} = \mathbf{r}_i^l + \phi_{vel}(\mathbf{r}^l, \mathbf{h}^l, i)$$

$$\phi_{vel}(\mathbf{r}^l, \mathbf{h}^l, i) = \begin{cases} \sum_{j \neq i} \frac{\mathbf{r}_i^l - \mathbf{r}_j^l}{d_{ij} + 1} \phi_r(\mathbf{h}_i^l, \mathbf{h}_j^l, d_{ij}^2) & \text{if node } i \text{ belongs to the point cloud } \mathbf{z}_t, \\ 0 & \text{if node } i \text{ belongs to the context } \mathbf{u} \end{cases}$$



Molecular Linker Design with Diffusion

Problem Statement

Given molecular fragments positioned in 3D space (fixed orientation), generate a linker that:

- Connects fragments at appropriate anchor atoms
- Has variable size (number of atoms)
- Avoids clashes with protein pocket
- Is chemically valid and synthesizable

Conditioning Strategies

Context u : Fragment coordinates + atom types

Linker center f : CoM of anchors or fragments

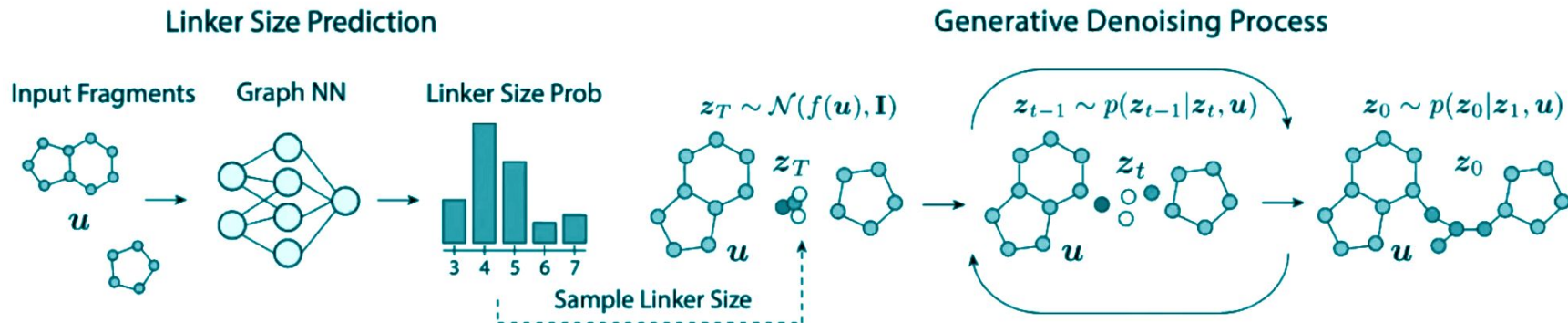
Pocket: Optional protein structure constraint

Two-Stage Process

Stage 1: Predict linker size using GNN

Stage 2: Generate linker atoms via conditioned diffusion

Equivariant Diffusion for Molecular Linker Design



Linker Size:

- Sampled from GNN probs
- Specified by user

Context u :

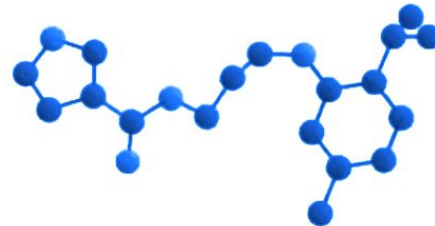
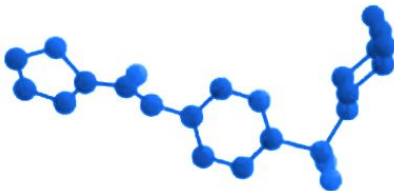
- Fragments (any number)
- Fragments + Pocket

Center f :

- CoM of anchors
- CoM of fragments

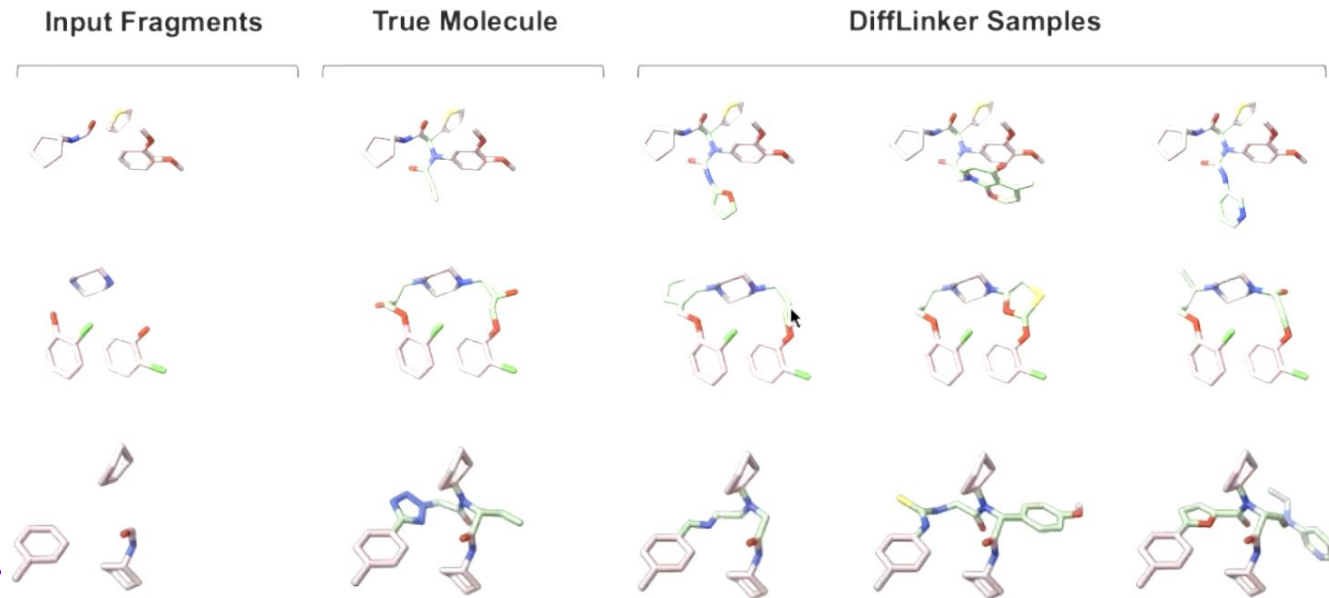
Linking Pairs of Fragments

	Method	QED $\uparrow$	SA $\downarrow$	# Rings $\uparrow$	Valid, %	Unique, %	Novel, %
ZINC	DeLinker + ConfVAE + MMFF	0.64	3.11	0.21	98.3	44.2	47.1
	3DLinker (given anchors)	0.65	3.11	0.23	99.3	29.0	41.2
	3DLinker	0.65	3.14	0.24	71.5	29.2	41.9
	DiffLinker	0.68	3.01	0.25	93.8	24.0	30.3
	DiffLinker (given anchors)	0.68	3.03	0.26	97.6	22.7	32.4
	DiffLinker (sampled size)	0.65	3.19	0.32	90.6	51.4	42.9
	DiffLinker (given anchors, sampled size)	0.65	3.24	0.36	94.8	50.9	47.7
CASF	DeLinker + ConfVAE + MMFF	0.35	4.05	0.35	95.7	51.6	55.6
	DiffLinker	0.41	4.00	0.34	85.3	40.5	41.8
	DiffLinker (given anchors)	0.40	4.03	0.38	90.2	37.3	48.4
	DiffLinker (sampled size)	0.40	4.06	0.30	63.7	60.0	49.3
	DiffLinker (given anchors, sampled size)	0.40	4.10	0.38	68.4	57.1	56.9



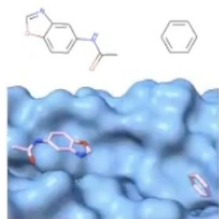
Linking >2 Fragments

Method		QED $\uparrow$	SA $\downarrow$	# Rings $\uparrow$	Valid, %	Unique, %	Novel, %
GEOM	3DLinker	0.36	3.56	0.00	16.3	73.7	—
	DiffLinker	0.48	2.99	0.75	94.0	34.7	68.6
	DiffLinker (given anchors)	0.49	3.01	0.79	94.0	35.0	68.4
	DiffLinker (sampled size)	0.45	3.27	0.75	87.2	62.4	76.0
	DiffLinker (given anchors, sampled size)	0.46	3.33	0.83	88.8	63.6	76.0

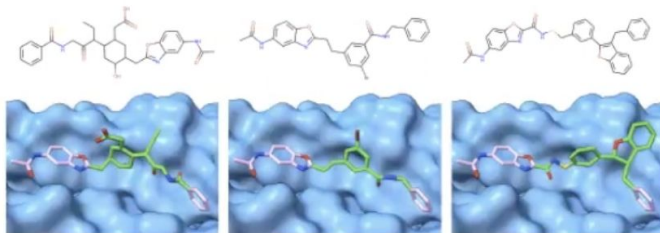


Pocket Conditioning

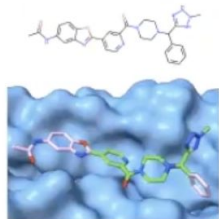
Input Fragments



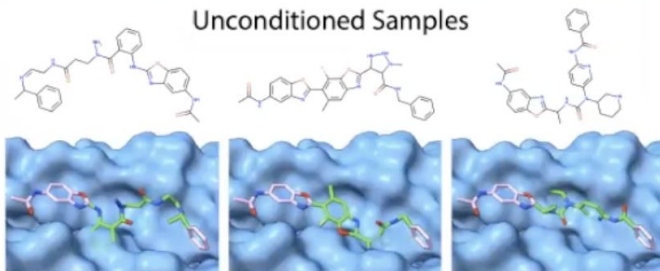
Pocket-Conditioned Samples



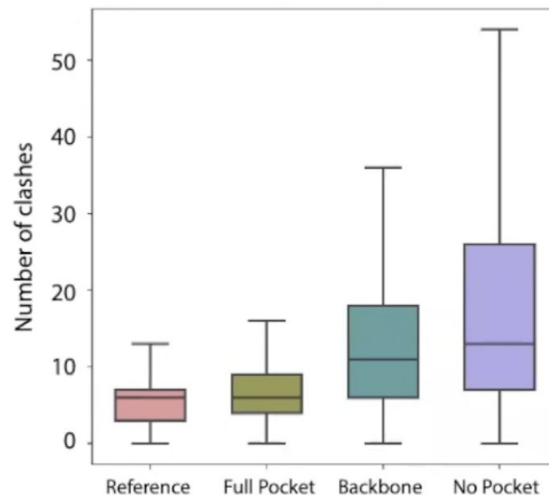
True Molecule



Unconditioned Samples



Numbers of clashes in DiffLinker samples with different pocket representations

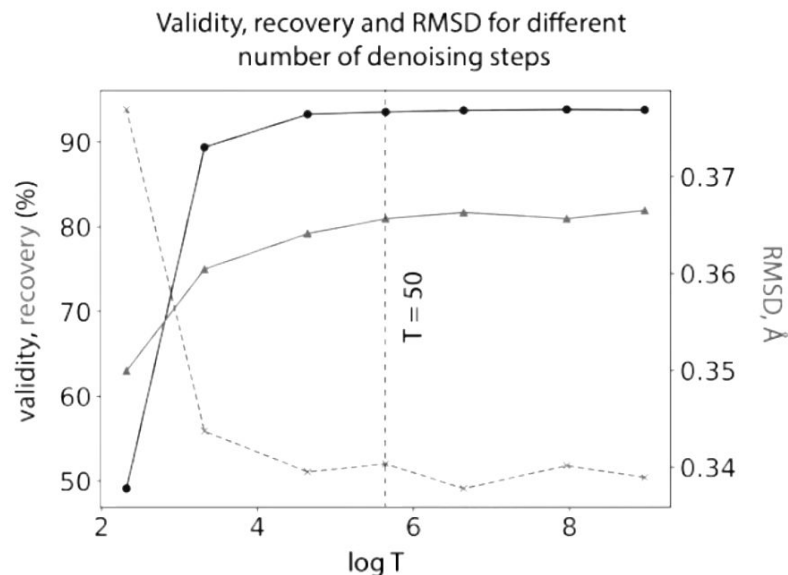


Method		QED $\uparrow$	SA $\downarrow$	# Rings $\uparrow$	Valid, %	Unique, %	Novel, %
Pockets	DiffLinker (pocket atoms)	0.45	3.89	1.06	88.7	62.5	73.1
	DiffLinker (pocket backbone)	0.45	3.77	0.95	90.4	60.9	72.8
	DiffLinker (unconditioned)	0.45	3.83	1.10	93.6	61.1	74.9

Performance

Sampling time for different datasets (with $T=500$).
Experiments were performed on a single Tesla
V100-PCIE-32GB GPU

dataset	batch size	time per molecule, s
ZINC	128	0.37
CASF	128	0.63
Multi-Fragment	64	0.37
Pockets (full atomic representation)	32	0.46
Pockets (backbone representation)	32	0.21
Pockets (no pocket)	32	0.12



DiffLinker is robust to significant reduction of the number of denoising steps in sampling resulting in 10-fold gain in sampling speed without any performance degradation

Transformer-based language models in CC

Table 1

Case studies: Transformer-Based Language Models for computational chemistry and cheminformatics.

Backbone	Encoder-based CLMs	Decoder-based CLMs	Encoder-decoder CLMs
Transformers(Attention)	BERT [28]	Guo et al. [29]	TransAntivirus (IUPAC) [30]
	MOLBERT [14]	GPT-2-based model [31]	Chemformer (BART-based) [32]
	SMILES-BERT [12]	MolXPT (GPT-2-based) [33]	MolT5 [34]
	ChemBERTa [13]	BioGPT [35]	Transformer-based models [36]
	ChemBerta-2 [37]	MolGPT (Relative Attention) [38]	X-MOL [39]
	GPT-MolBERTa [40]	Mol-Instructions [41]	SMILES transformer [42]
	RoBERTa-based model [43]		Transformer Neural Network [44]
	MolFormer [45]		Transformer-based ANN [46]
	SELMFormer (SELFIES) [47]		RT (SELFIES) [48]
	Mol-BERT [49]		
	MolRoPE-BERT [50]		
	BET [51]		

Chemical Language Models

SMILES Notation

Simplified Molecular Input Line Entry System: A text-based representation of molecular structures

Example (Aspirin):

```
CC(=O)Oc1ccccc1C(=O)O
```

Atom-Level

Each character is a token

```
C, C, (, =, O, ...
```

k-mers (n-grams)

Fixed-length substrings

```
CC(, C(=, (=O, ...
```

BPE/SPE

Data-driven tokenization

```
CC, (=O), Oc, ...
```

Transformer Models for Chemistry

Encoder-based:

BERT, ChemBERTa, MolBERT

Property prediction

Decoder-based:

MolGPT, BioGPT

Generation

Encoder-Decoder:

BART, T5, MolT5

Translation tasks

MolGPT: Molecular Generation

Architecture Overview

GPT-style decoder model conditioned on molecular scaffolds and desired properties

Input Representation

Segment Tokens: Distinguish condition (0) from generated sequence (1)

Position Tokens: Standard positional encoding

Condition: Scaffold SMILES + properties (LogP, TPSA, etc.)

Decoder Blocks

Masked Self-Attention: Causal masking for autoregressive generation

Feed-Forward Network: Position-wise transformations

Layer Normalization: Stabilize training

Training Process

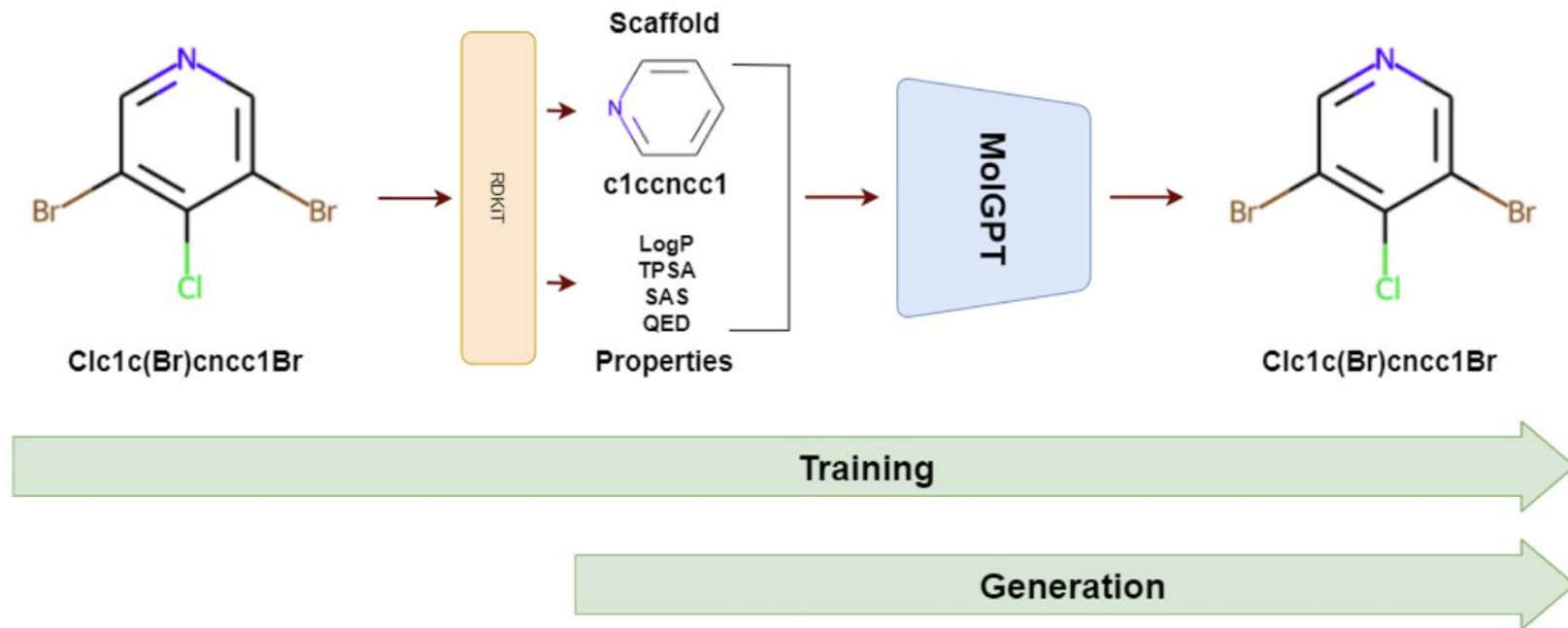
Step 1: Extract scaffold from molecule using RDKit

Step 2: Compute properties (LogP, TPSA, SAS, QED)

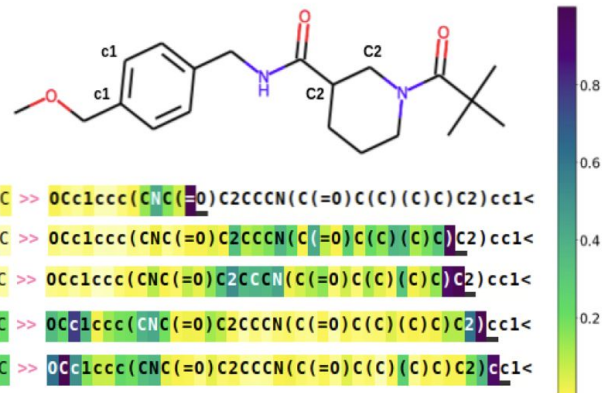
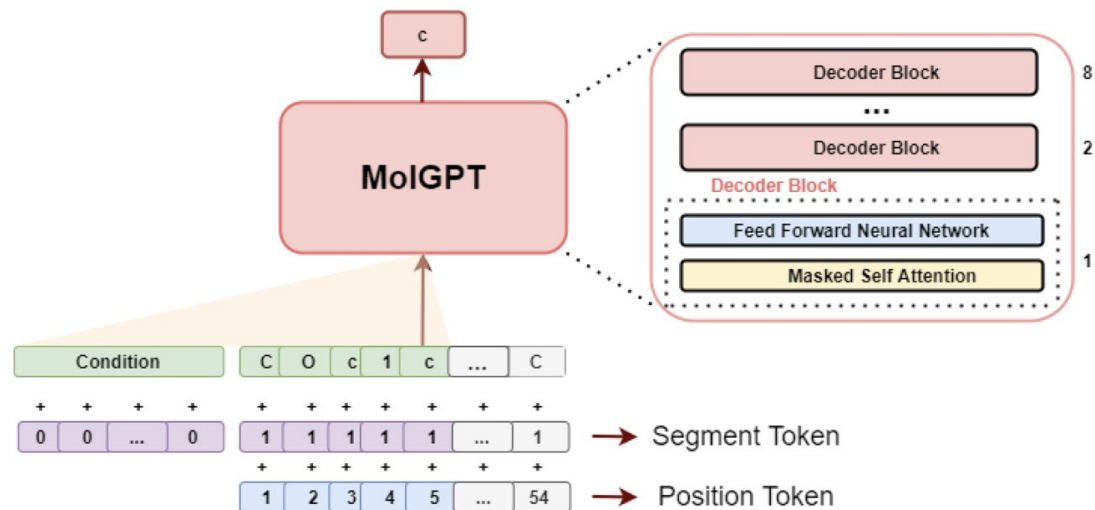
Step 3: Train to predict next token given scaffold + properties + previous tokens

```
Loss = CrossEntropy(predicted_token, actual_token)
```


MolGPT Training



MolGPT Model Architecture



Reinforcement Learning for Molecule Optimization

RL Framework for Drug Discovery

Use RL to optimize pre-trained generative models toward molecules with specific desired properties

RL Components

State: Partial SMILES sequence

Action: Next token to append

Agent: Language model (MolGPT, etc.)

Reward: Computed at end of episode

Reward Functions

Validity: Is it a valid molecule?

Drug-likeness: QED, Lipinski's rules

Target property: Binding affinity, solubility

Diversity: Penalize similar molecules

Two-Stage Training

Stage 1: Pre-training

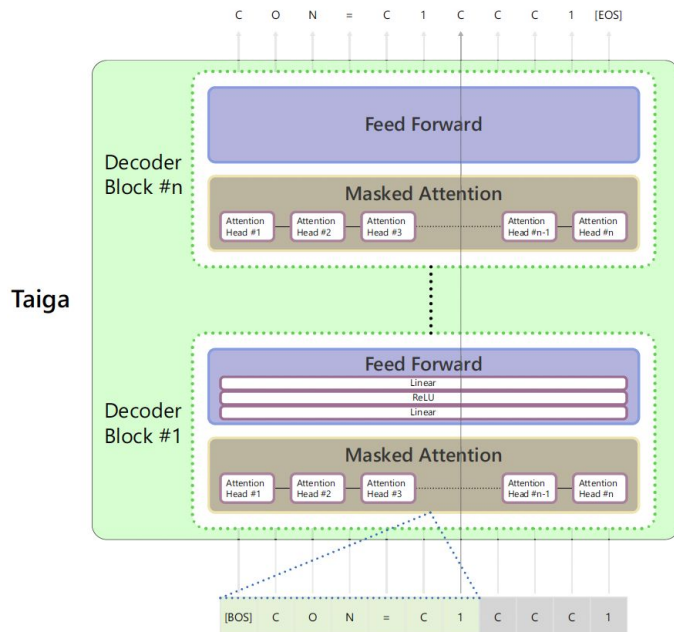
Learn chemical syntax and validity from large corpus (e.g., 10M SMILES from PubChem)

Stage 2: RL Fine-tuning

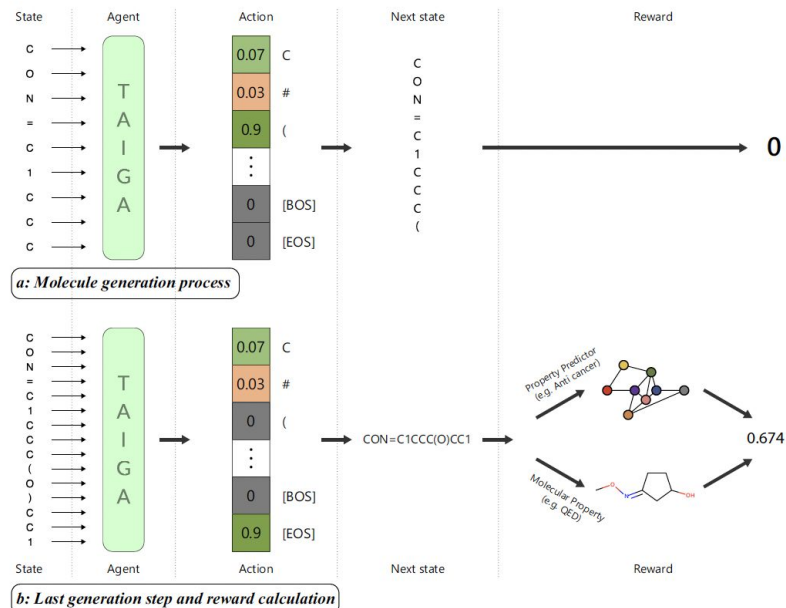
Optimize for target properties using policy gradient methods (REINFORCE, PPO)

Challenge: Balance exploitation (optimizing reward) vs. exploration (maintaining diversity). Solution: Add KL divergence penalty to keep close to pre-trained distribution.

Reinforcement learning



Stage 1



Stage 2

Multimodal AI for Drug Discovery

TamGen Architecture

1. Protein Encoder: Process 3D structure and amino acid sequence with geometric GNN

2. Cross-Attention: Attend to protein features while generating SMILES

3. Chemical Decoder: GPT-like model conditioned on protein representation

Training Data

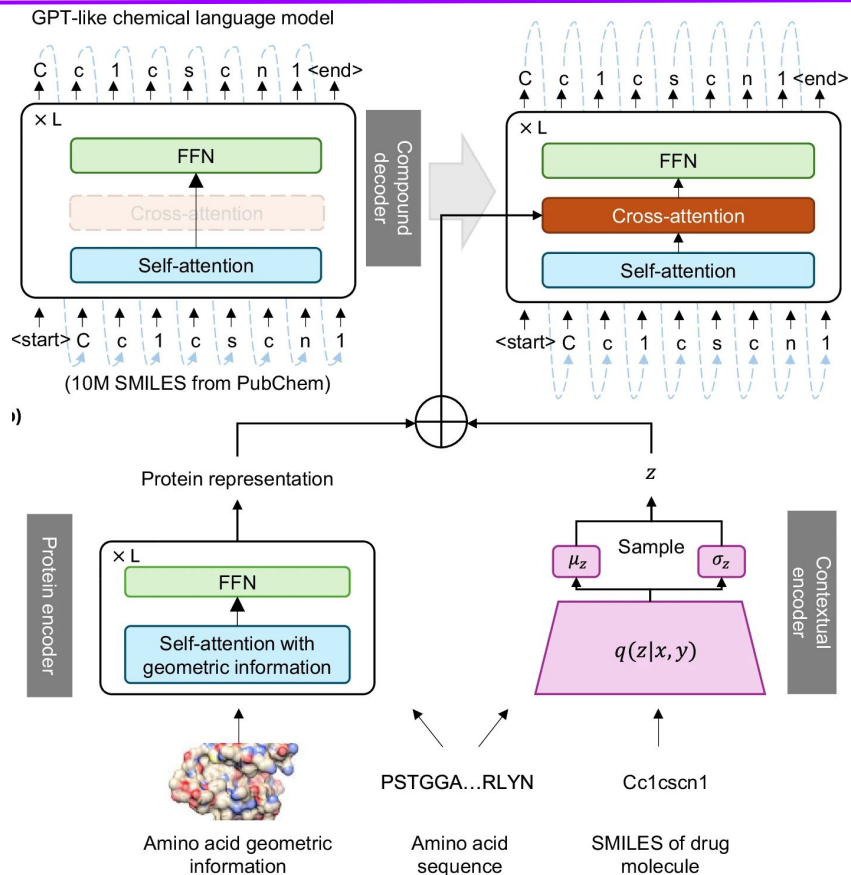
- 10M SMILES (pre-training)
- Protein-ligand complexes from PDB
- Binding affinity data (BindingDB)
- Multi-task learning objective

Evaluation Metrics

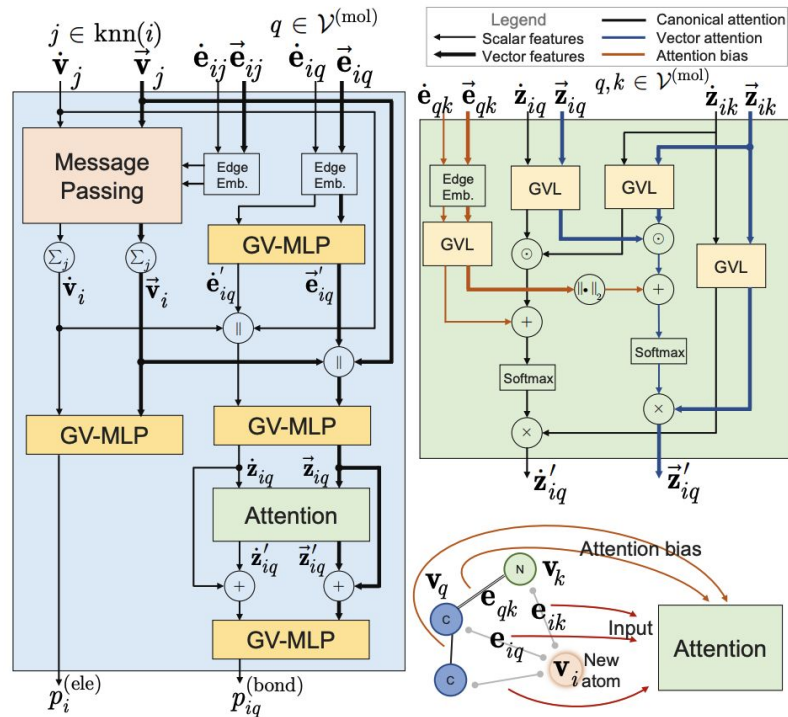
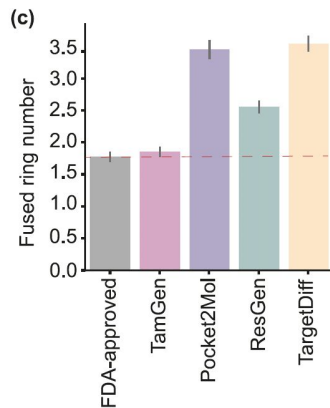
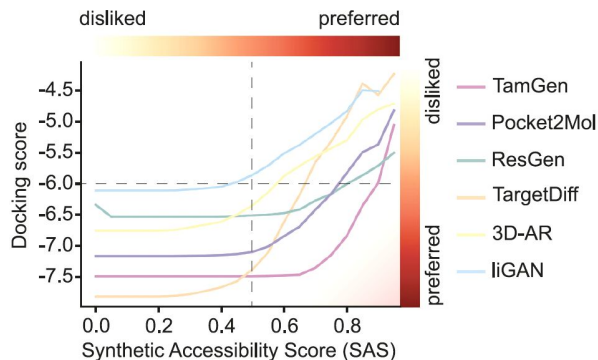
- Docking score (QED)
 - Synthetic accessibility (SAS)
 - Diversity & novelty
 - Experimental validation (IC₅₀)
-

Unlocking Drug Discovery with Multimodal AI

- **Current challenge:** AI-generated drug candidates often have limited real-world impact due to isolated ligand-based or structure-based approaches, leading to poor outcomes in subsequent discovery stages.
- **Proposed solution:** Leveraging multimodal AI—specifically, a GPT-like chemical language model—for target-aware molecule generation and refinement.
- **Key results:** TamGen generated higher-quality, viable compounds, successfully identifying 5 promising candidates against a tuberculosis protease, with the strongest inhibitor achieving an IC_{50} of 1.9 μM .



Compound generation performance



Wu et al., Nat. Comm., 2024

Peng et al., arXiv, 2022

Molecular Docking & Scoring

What is Molecular Docking?

Computational method to predict the preferred orientation and binding affinity of a small molecule (ligand) to a protein target, simulating how a drug candidate might interact with its target.

Docking Process

- 1. Pose Generation:** Sample different orientations
- 2. Scoring:** Evaluate binding energy for each pose
- 3. Optimization:** Refine best poses

Scoring Functions

- Force-field:** Physics-based (AMBER, CHARMM)
- Empirical:** Trained on experimental data
- Knowledge-based:** Statistical potentials
- ML-based:** Neural networks (recent)

Interpreting Docking Scores

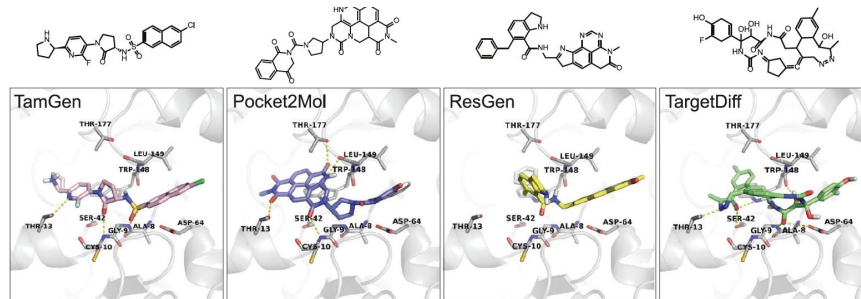
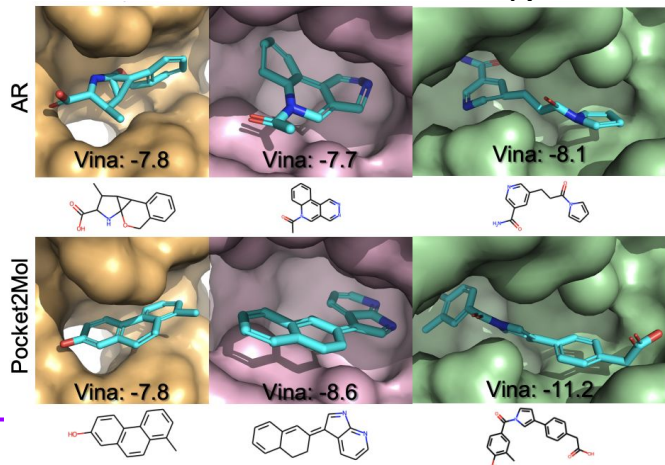
Units: Typically kcal/mol (negative = favorable binding)

- Score < -7: Weak binding
- Score -7 to -9: Moderate binding
- Score < -9: Strong binding

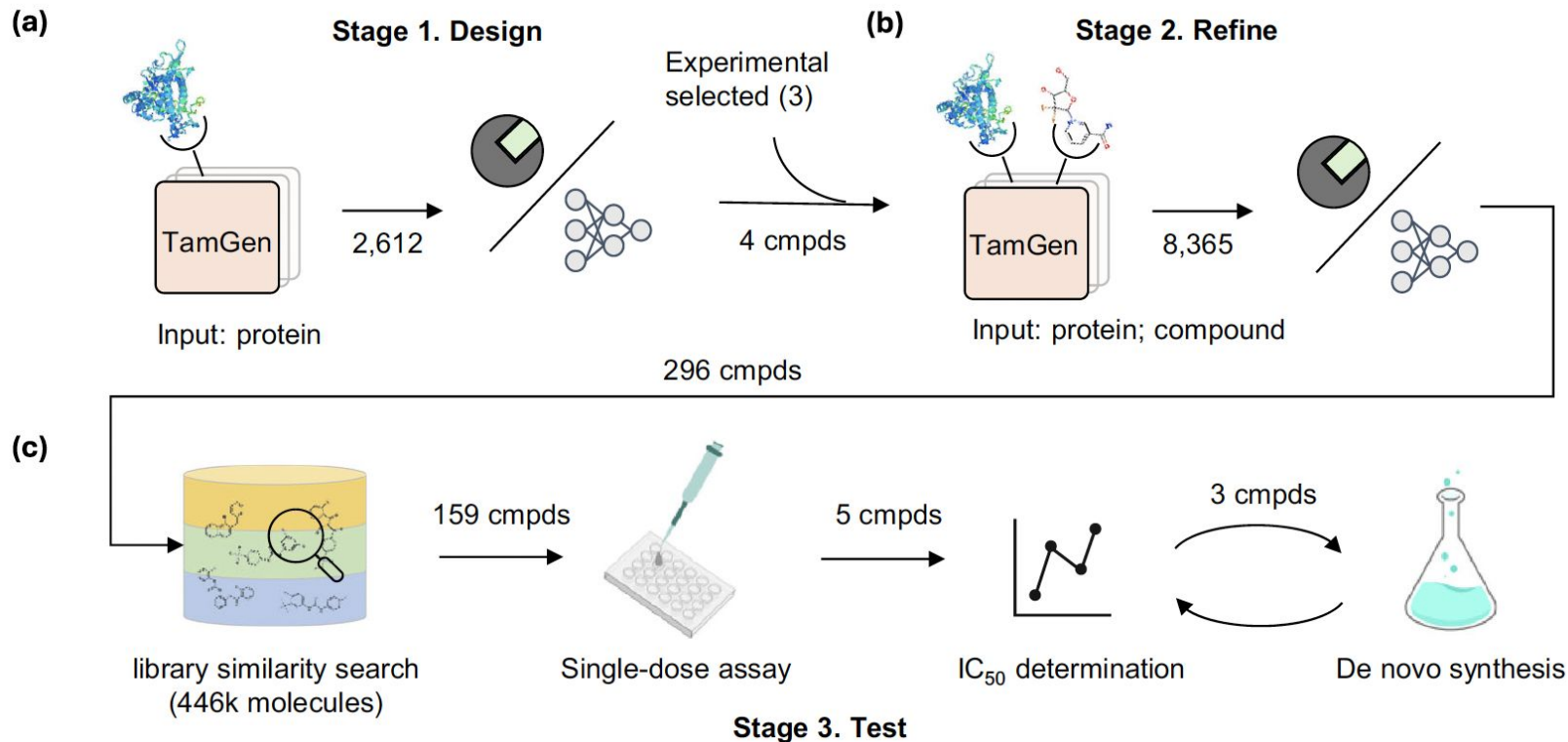
Note: Scores are heuristic estimates. Experimental validation (IC_{50} , K_d) is essential.

Docking Visualization

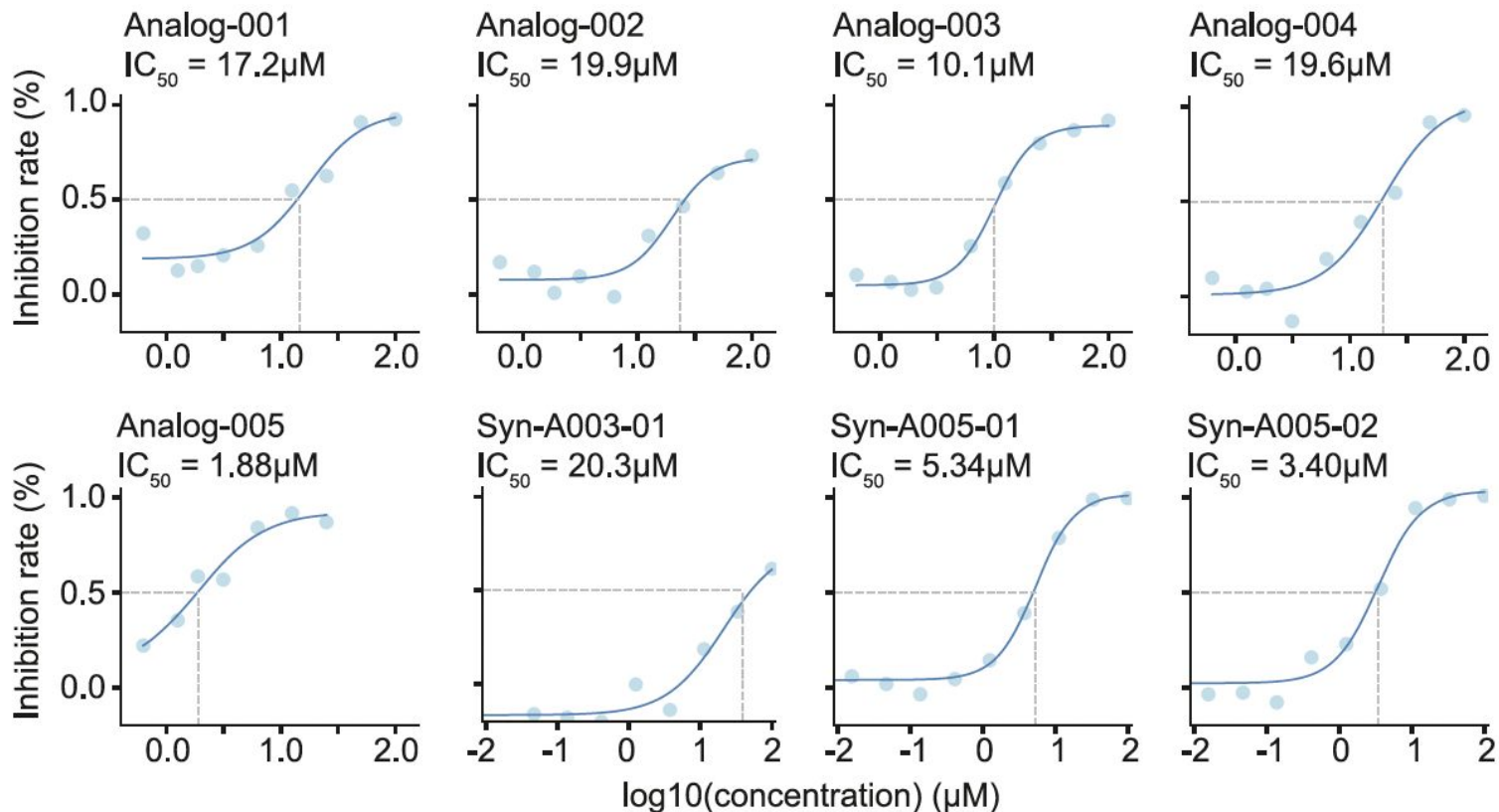
- The **protein** is the **target** — say, a viral enzyme or a cancer-related protein.
- The **ligand** is a potential drug molecule that you want to **bind to the protein** and block or activate it.
- Docking software runs simulations to explore how the **ligand can interact with the protein's binding site**.
- The docking score is a heuristic or energy-based value (usually in kcal/mol) that estimates the binding affinity



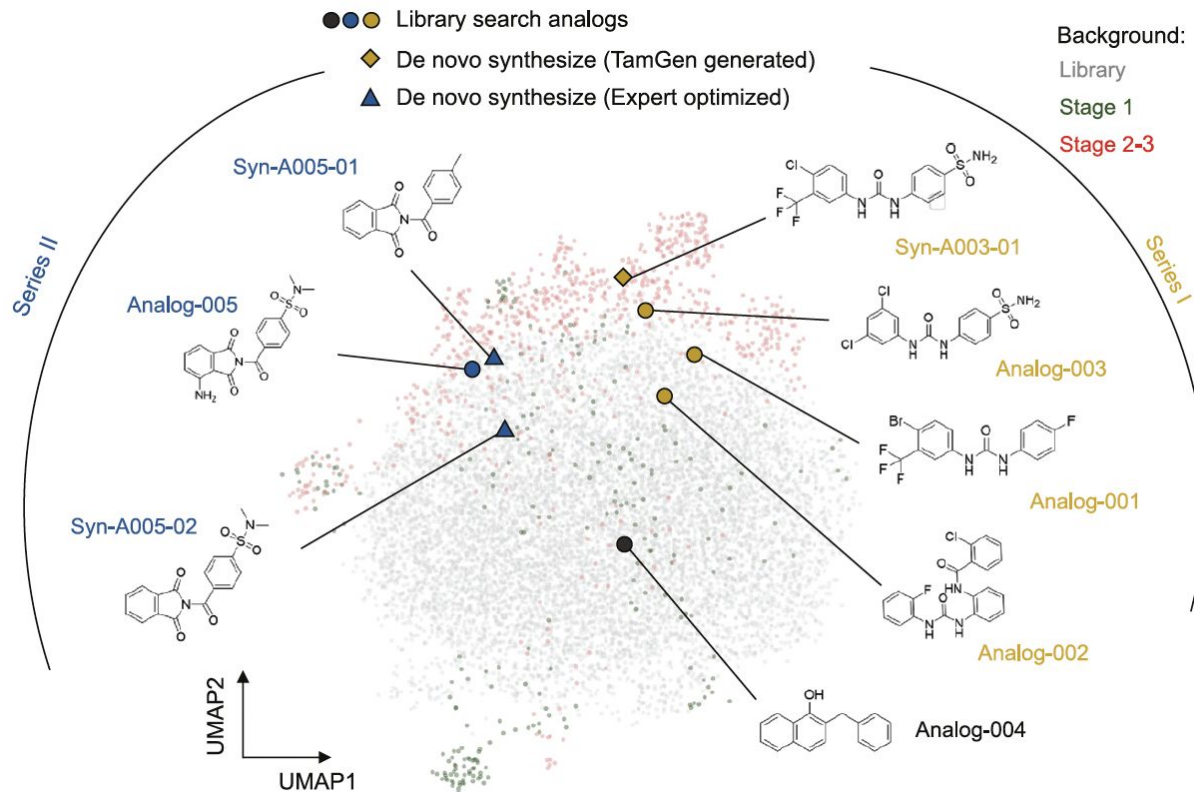
Design-Refine-Test Pipeline



Experimental validation of designed compounds

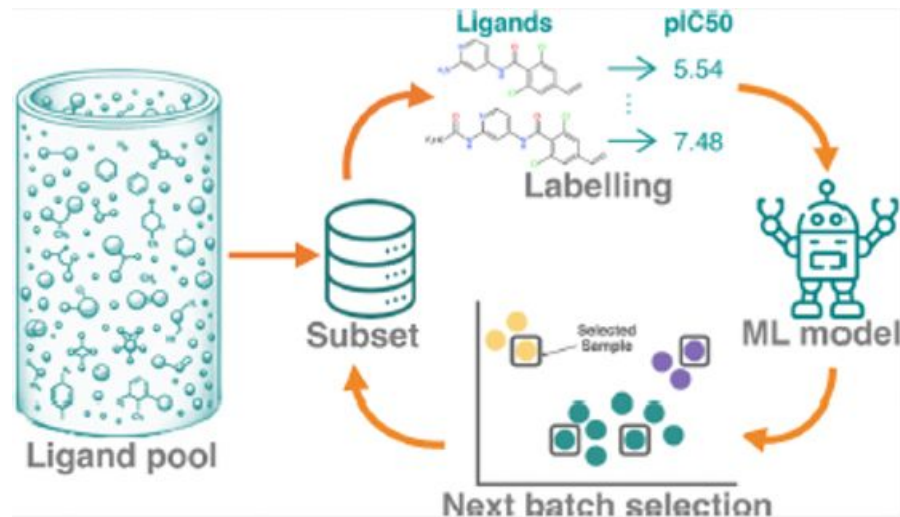


Visualization of chemical space



Active learning in drug discovery

- **Active learning:** A machine learning approach focused on selective data labeling
- The algorithm queries **the most informative data points** from a pool of unlabeled data
- Reduce experimental efforts and computational resources for labeling
- Infer an accurate ML model with a **smaller, curated training dataset**
- Achieve or surpass the accuracy of models trained on larger, randomly selected datasets
- **Iteratively select data points with the highest uncertainty** or potential impact on model performance



Active Learning in Drug Discovery

Active Learning Cycle

- 1 Train ML model on labeled data
- 2 Use model to score unlabeled library
- 3 Select most informative molecules (acquisition function)
- 4 Label selected molecules (simulation or experiment)
- 5 Add to training set and repeat

Acquisition Strategies

- **Uncertainty:** High prediction variance
- **Expected Improvement:** Likely to exceed current best
- **Upper Confidence Bound:** Balance mean + uncertainty
- **Diversity:** Cover chemical space

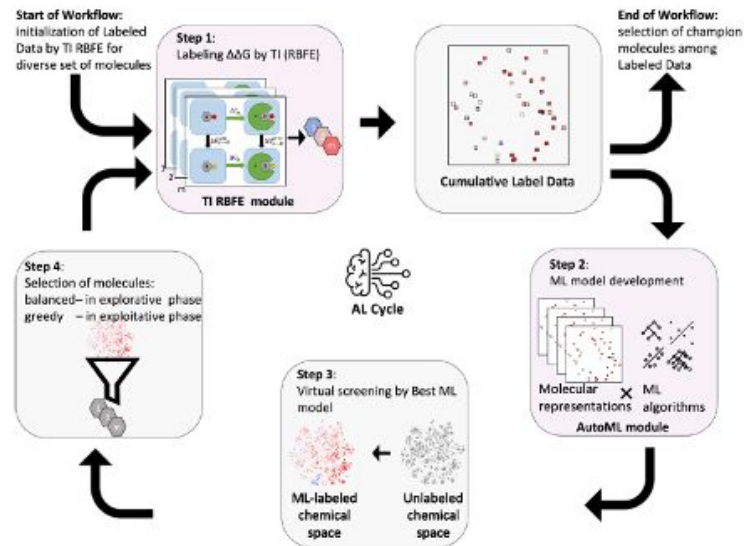
Application: RBFE Calculation

Thermodynamic integration (TI) is accurate but expensive (~hours per molecule). AL selects which molecules to compute with TI, achieving 100× affinity improvement with 20× speedup vs. random selection.

A typical AL pipeline

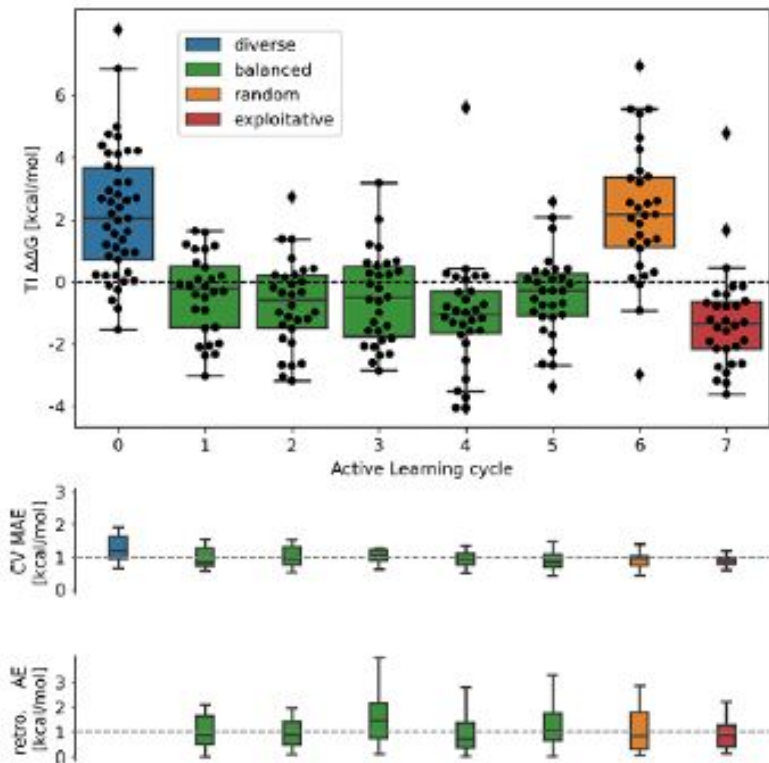
In silico ligand binding free energy calculation for drug discovery

- Finding the lowest **relative binding free energy (RBE)** among congeneric ligands
- **Thermodynamic integration (TI)** with molecular dynamics (MD) is highly effective but costly and time-intensive
- **AutoML and Active Learning (AL) integration** compounds for TI calculations are selected with ML to (1) enrich a set of molecules (exploitation), (2) improve an ML model's prediction of RBE for an entire library
- Achieves a **100-fold increase in binding affinity** and a 20-fold speedup compared to traditional brute force methods



AI cycles of explorative and exploitative regimes

- **Exploitative** and **explorative** regimes are distinguished by **greedy selection vs balanced selection**.
- Explorative regime was used until convergence in retrospective AE.
- With AL progress, ML model's performance improves, achieving a **cross-validated MAE of 1 kcal/mol**, comparable to RBF accuracy.
- The process converged effectively, with a significant improvement in mean TI $\Delta\Delta G$ during the **exploitative AL cycle 7** compared to the random selection in cycle 6.



M.D.

Applying Active Learning in Drug Discovery



RELAY
LABORATORIES

Google



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